

SLCO4A - Aula Prática 4

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Sinais

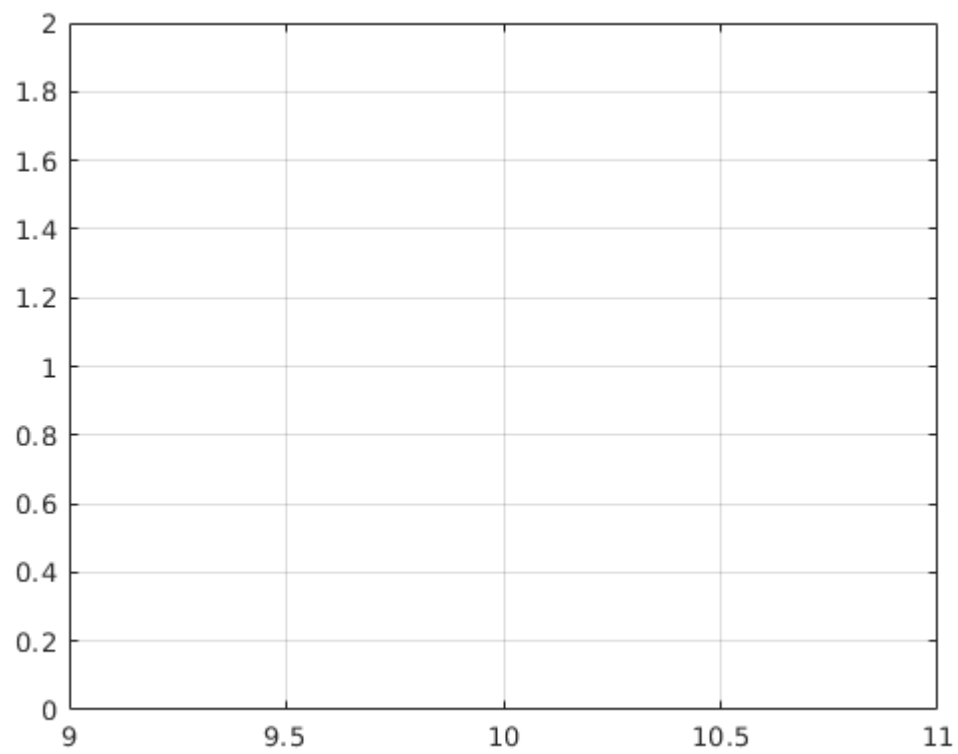
Representação de sinais

Sinal degrau

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

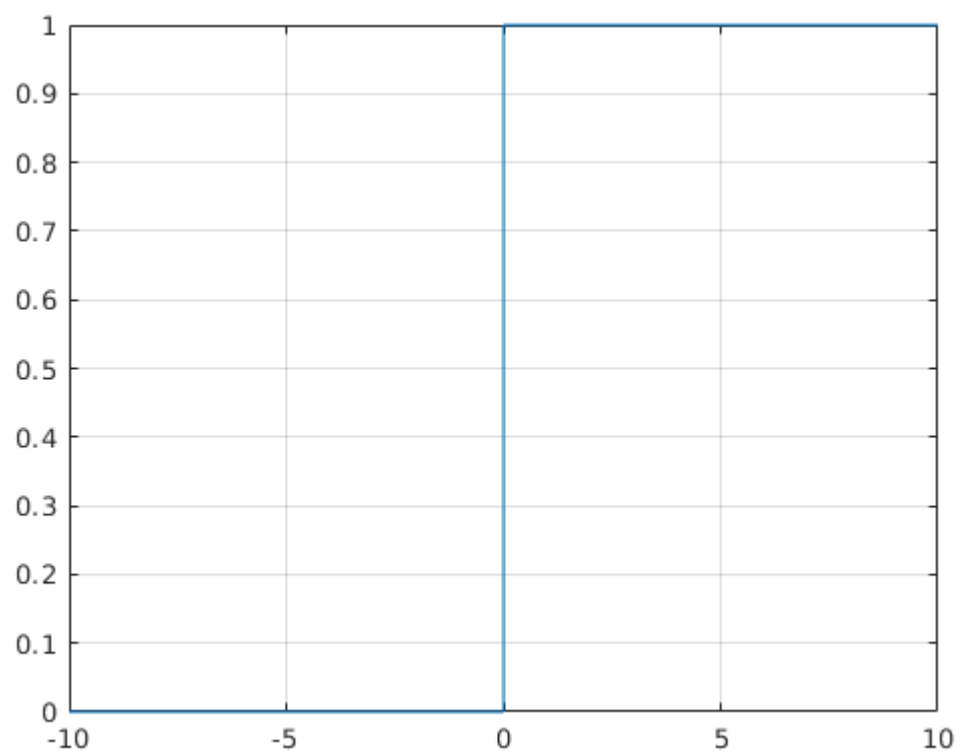
Primeiro método

```
t=10:0.1:10;  
u=heaviside(t);  
plot(t,u)  
grid on
```



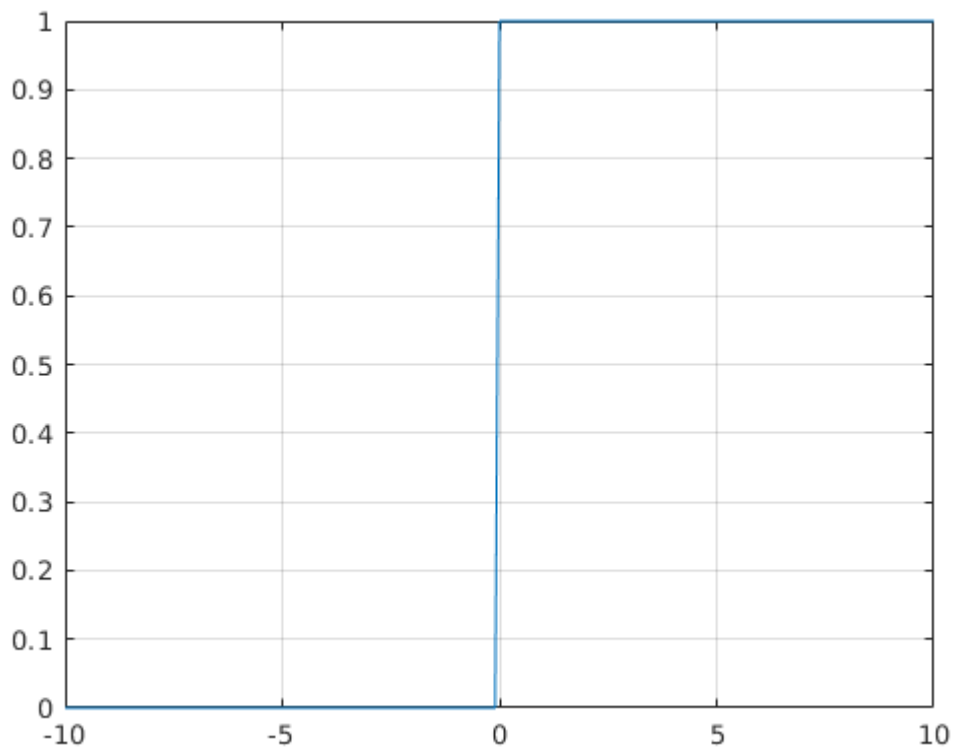
Segundo método

```
t1=-10:0;1:-0.1;  
t2=0:0.1:10;  
u1=zeros(size(t1));  
u2=ones(size(t2));  
t=[t1 t2];  
u=[u1 u2];  
plot(t,u)  
grid on
```



Terceiro método

```
t=-10:0.1:10;  
uspet=t>=0;  
plot(t,uspet)  
grid on
```

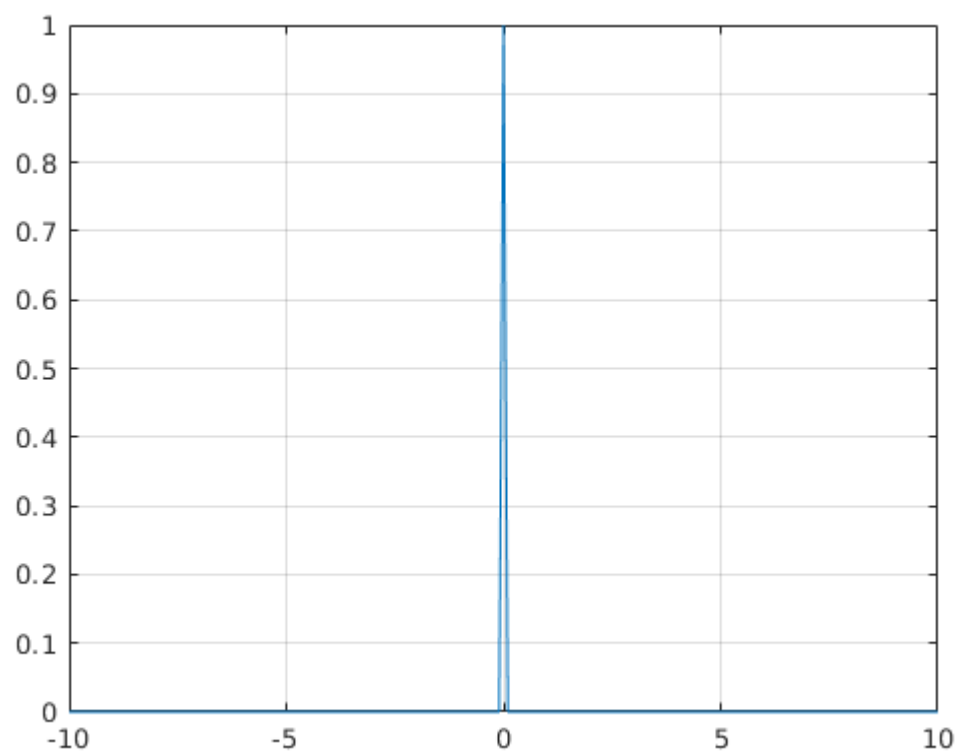


Sinal impulso unitário ou função delta de Dirac $\delta(t)$

$\int_{-\infty}^{\infty} f(t)\delta(t)dt = 1$, se $f(t) = 1, \forall t \in \mathfrak{R}$ então $\int_{-\infty}^{\infty} \delta(t) = 1$

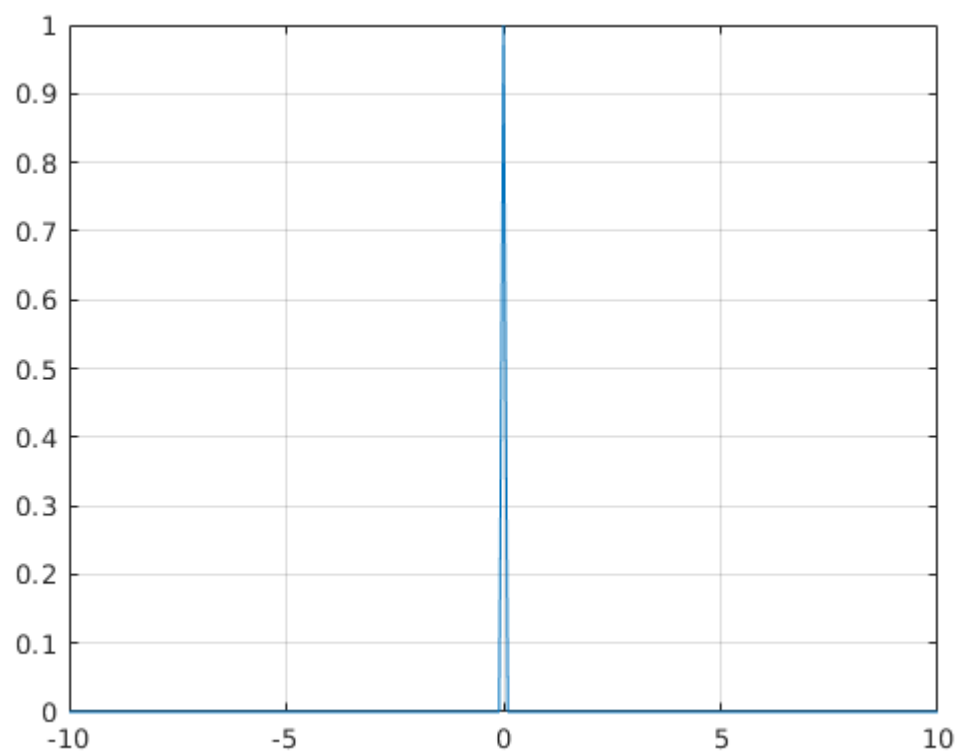
Primeiro método

```
t1=-10:0.1:-0.1;
t2=0;
t3=0.1:0.1:10;
d1=zeros(size(t1));
d2=1;
d3=zeros(size(t3));
t=[t1 t2 t3];
d=[d1 d2 d3];
plot(t,d)
grid on
```



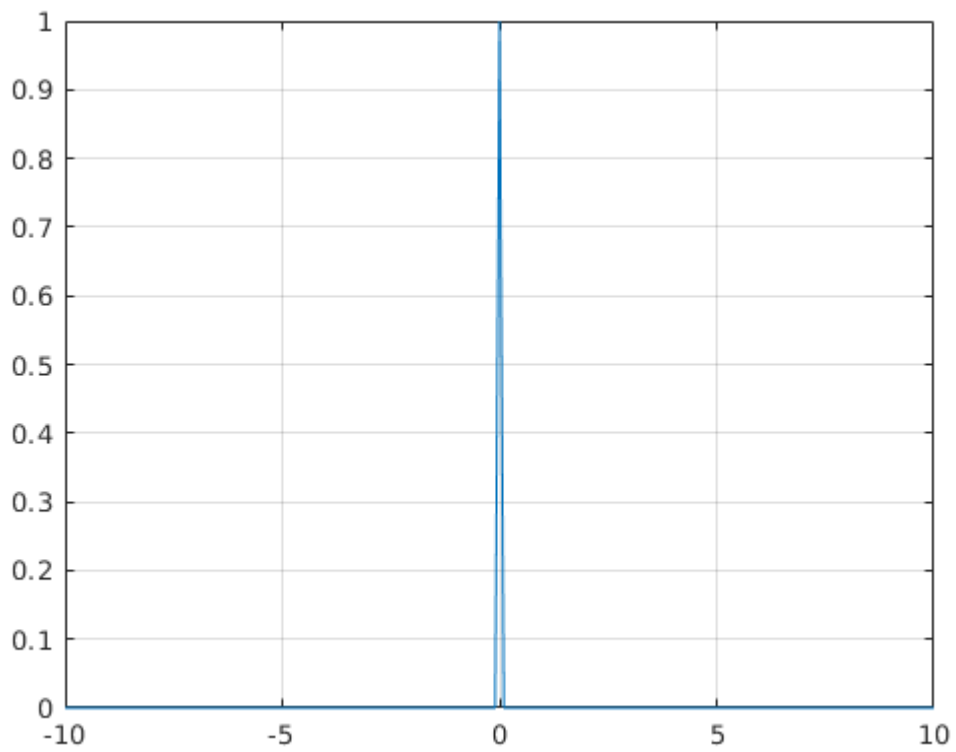
Segundo método

```
t1=-10:0.1:10;  
d=t==0;  
plot(t,d)  
grid on
```



Terceiro método

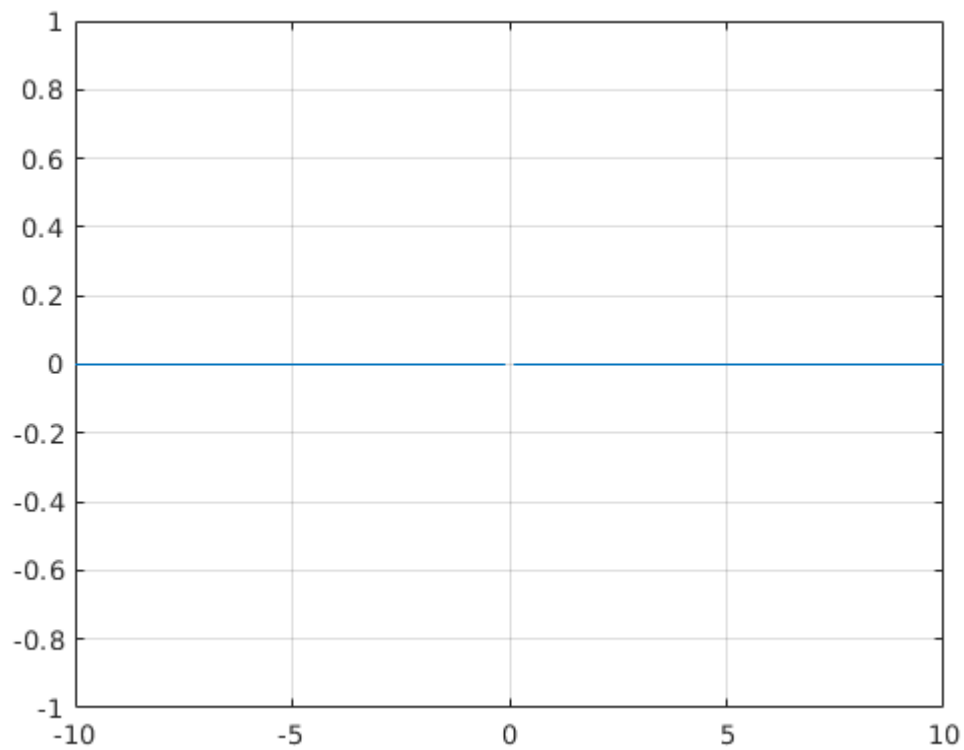
```
t=-10:0.1:10;  
d=gauspuls(t);  
plot(t,d)  
grid on
```



Quarto método

$$\delta(t) = \begin{cases} \infty & t = 0 \\ 0 & t \neq 0 \end{cases}$$

```
t=-10:0.1:10;  
d=dirac(t);  
plot(t,d)  
grid on
```



Propriedade da integral

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

```
syms t
d=dirac(t)
```

```
d =  $\delta(t)$ 
```

```
res=int(d,t,-inf,inf)
```

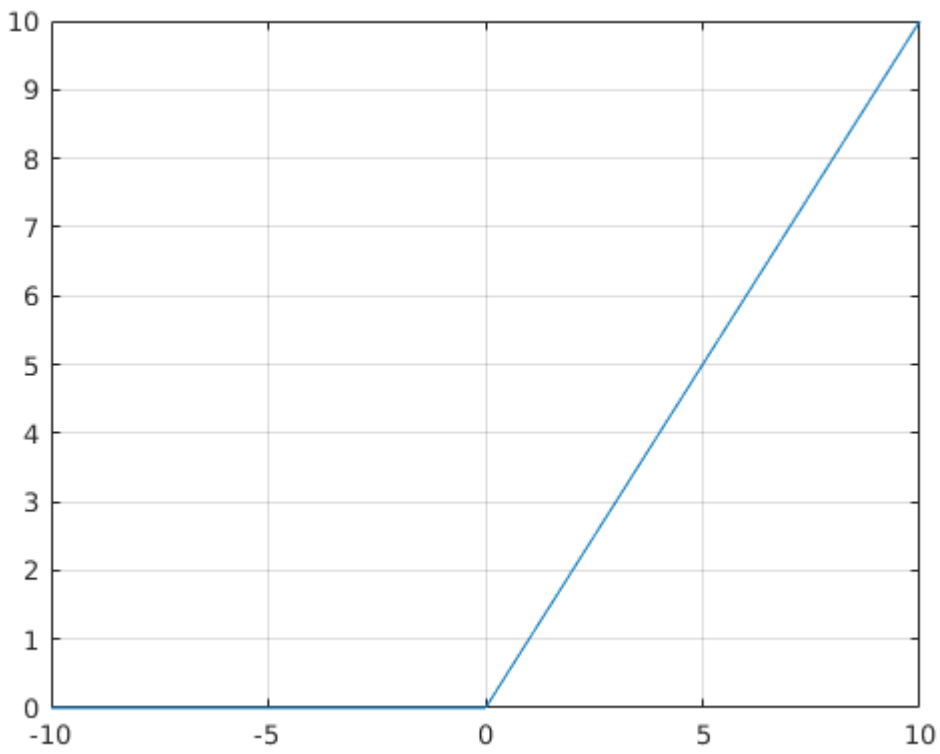
```
res = 1
```

Sinal rampa

$$r(t) = t \cdot u(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

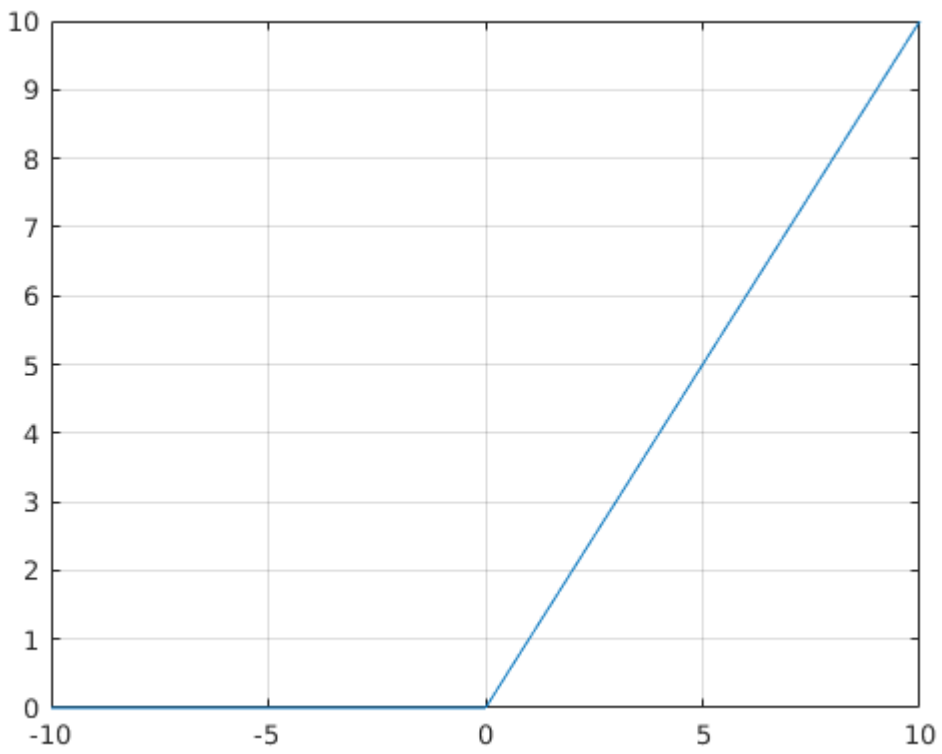
Primeiro método

```
t=-10:0.1:10;
r=t.*heaviside(t);
plot(t,r)
grid on
```

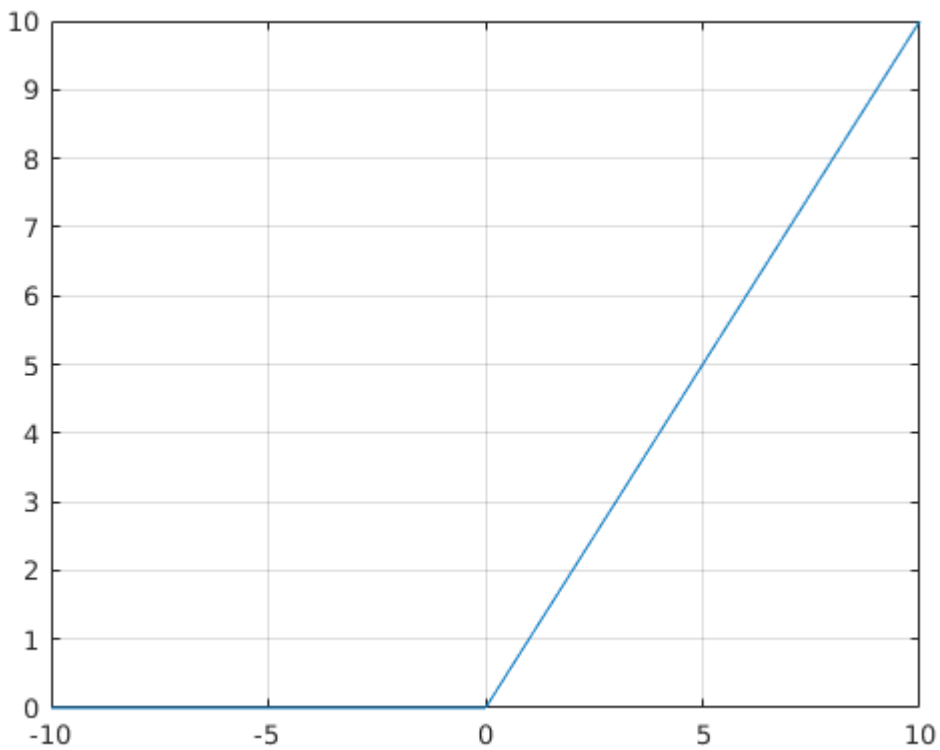
Segundo método

```
t1=-10:0.1:-0.1;  
t2=0:0.1:10;  
r1=zeros(size(t1));  
r2=t2;  
t=[t1 t2];  
r=[r1 r2];  
plot(t,r)  
grid on
```



Terceiro método

```
t=-10:0.1:10;  
r=(t>=0).*t;  
plot(t,r)  
grid on
```



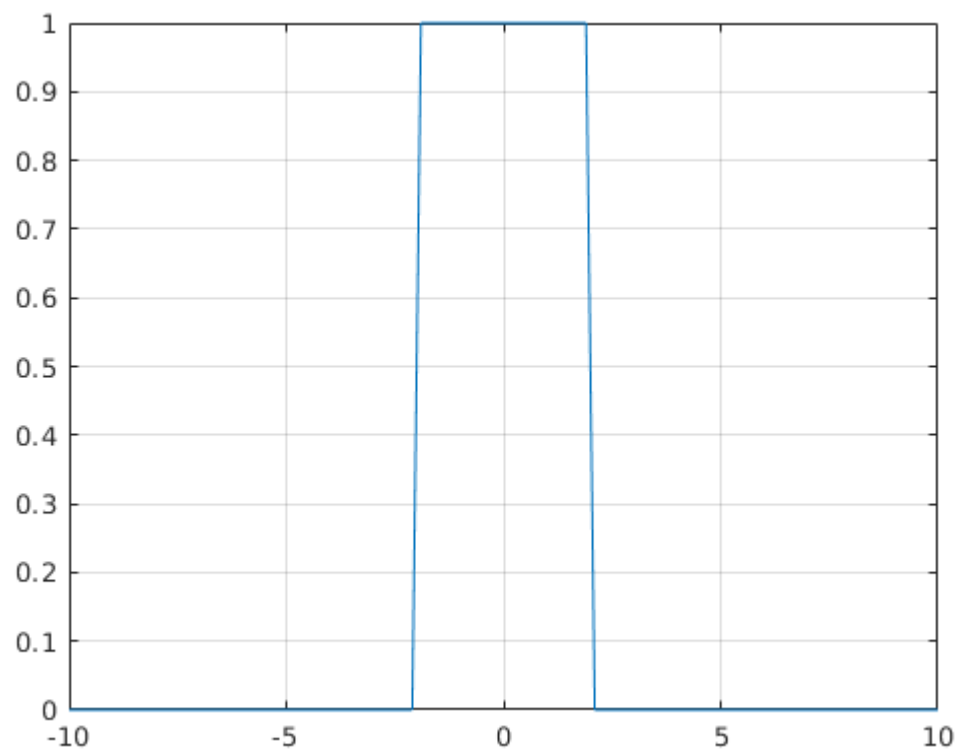
Sinal pulso retangular

$$p_T(t) = u\left(t + \frac{T}{2}\right) - u\left(t - \frac{T}{2}\right) = \begin{cases} 1 & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0 & \text{cc} \end{cases}$$

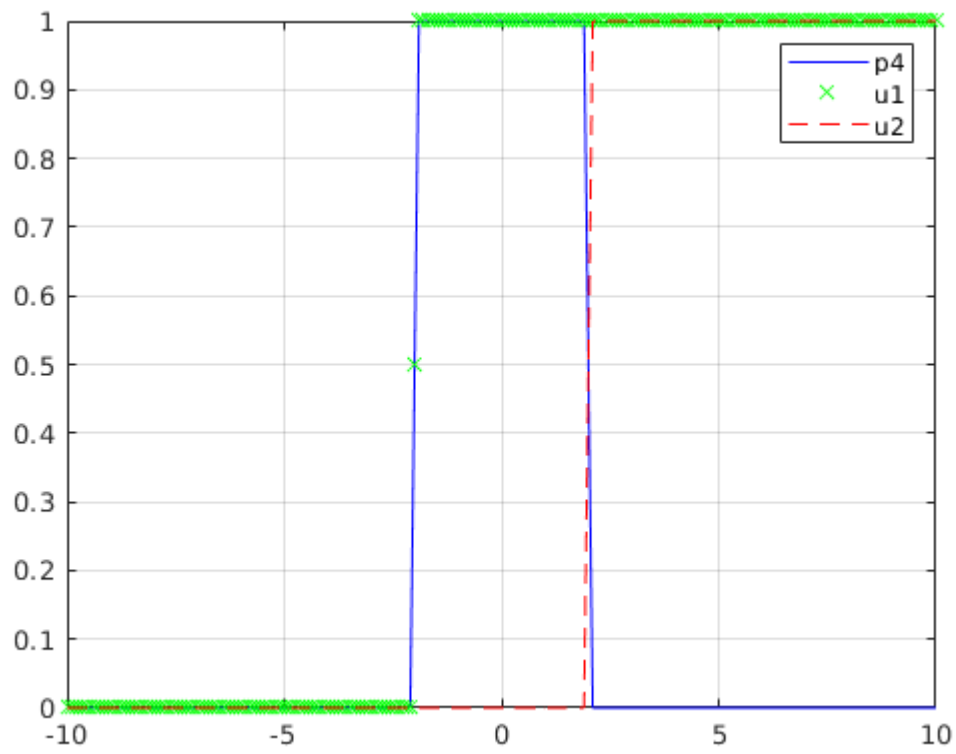
a) Considere $T = 4$, obtenha a representação gráfica de $p_4(t)$.

Primeiro método

```
t=-10:0.1:10;
p4=heaviside(t+2)-heaviside(t-2);
plot(t,p4)
grid on
```

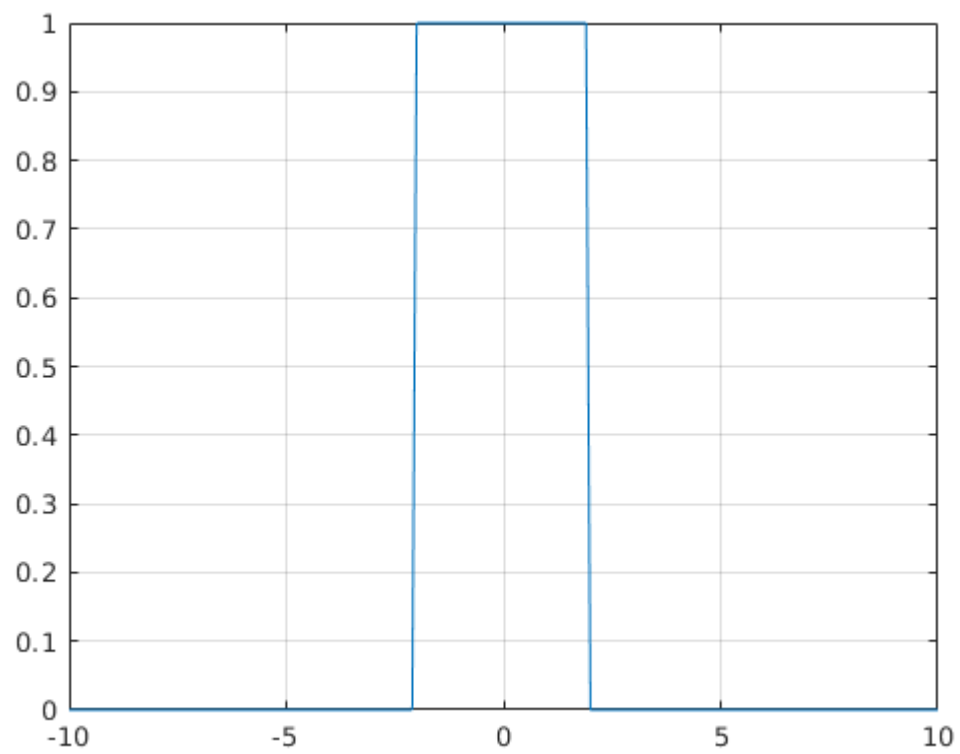


```
u1=heaviside(t+2);  
u2=heaviside(t-2);  
p4=u1-u2;  
plot(t,p4, 'b')  
hold on  
plot(t,u1, 'gx', t,u2, 'r--')  
grid on  
legend('p4', 'u1', 'u2')
```



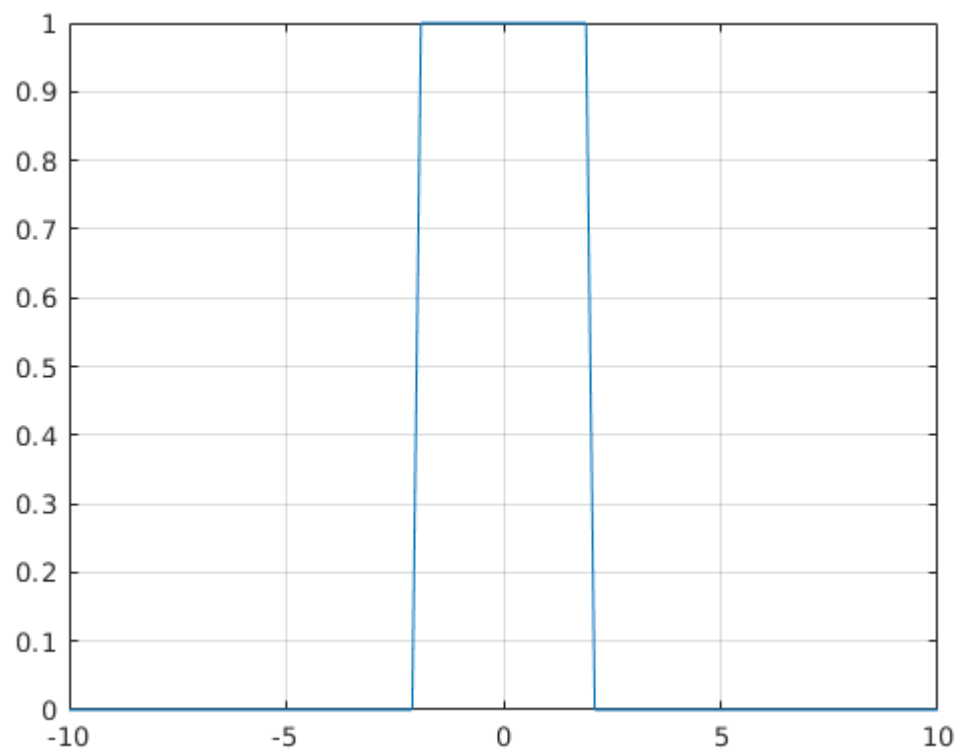
Segundo método

```
t=-10:0.1:10;  
s=rectpuls(t,4);  
figure  
plot(t,s)  
grid on
```



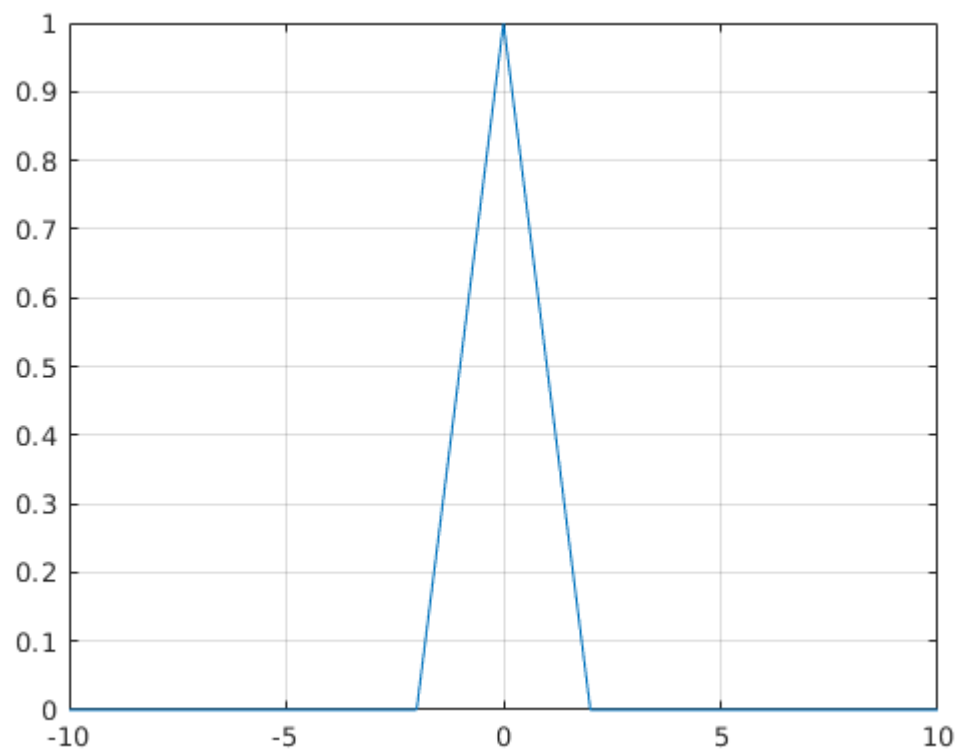
Terceiro método

```
t=-10:0.1:10;  
s=rectangularPulse(-2,2,t);  
plot(t,s)  
grid on
```



Sinal triangular

```
t=-10:0.1:10;  
tri=triangularPulse(-2,0,2,t);  
plot(t,tri);  
grid on
```

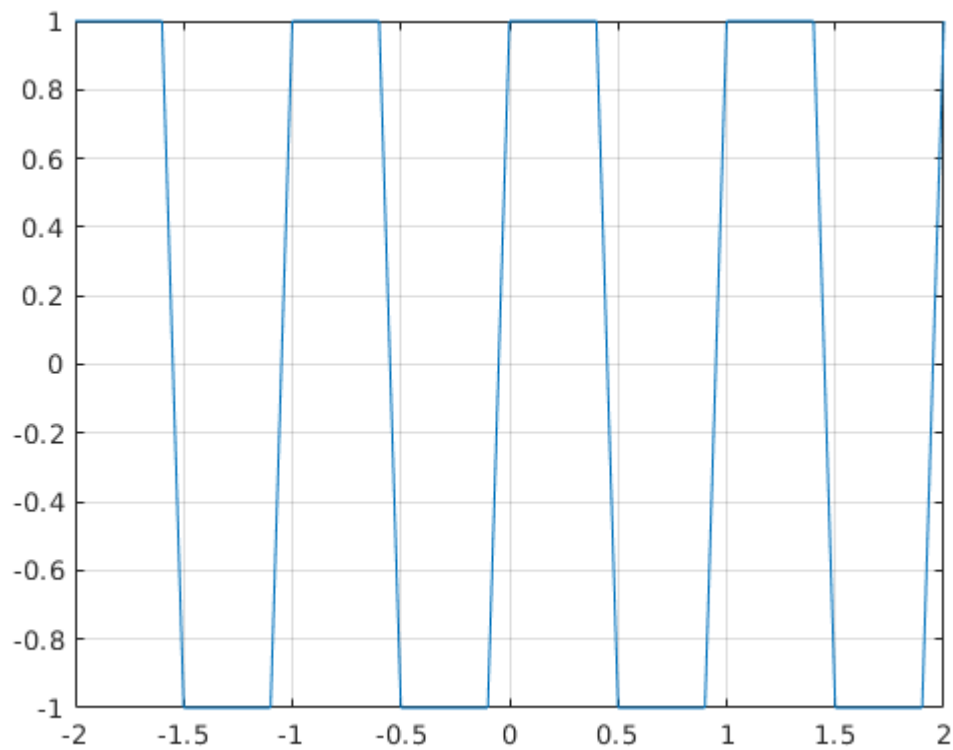


Sinal onda quadrada

```
t=-2:0.1:2;  
s=square(2*pi*t)
```

```
s = 1×41  
    1     1     1     1     1    -1    -1    -1    -1    -1     1     1     1 ...
```

```
plot(t,s)  
grid on
```

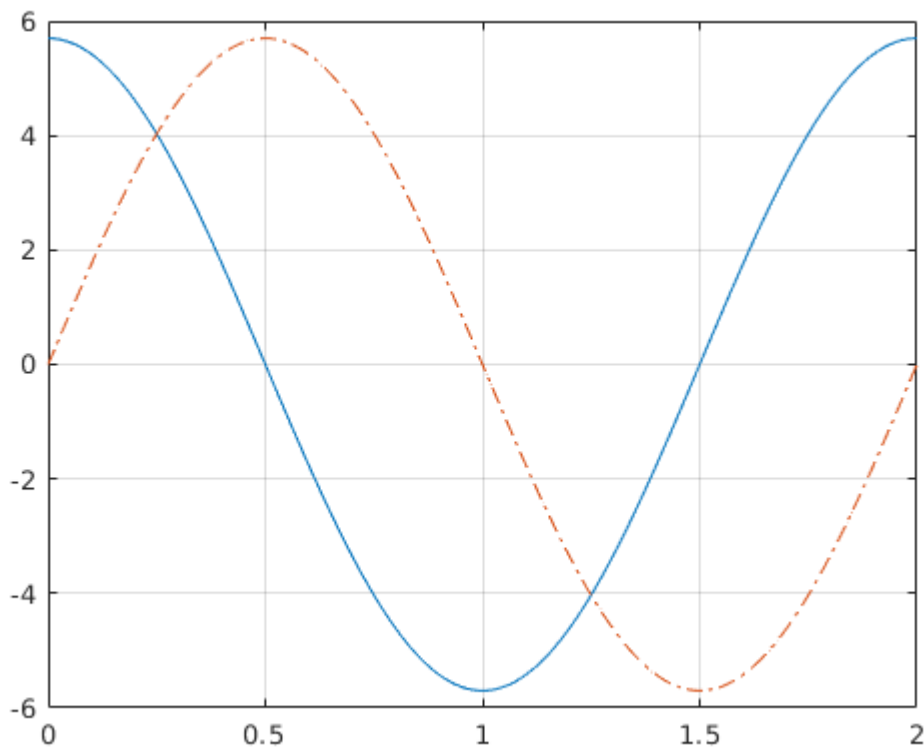



Sinal exponencial complexa

$$Ae^{j\Omega t + \theta} = A\cos(\Omega t + \theta) + j\sin(\Omega t + \theta)$$

a) Obtenha a representação da parte real e imaginária do sinal $y(t) = 2e^{j\pi t + \frac{\pi}{3}}$ no intervalo de um período

```
T=2;
t=0:0.01:T;
y_re=real(2*exp(j*pi*t+pi/3));
y_img=imag(2*exp(j*pi*t+pi/3));
plot(t,y_re,t,y_img, '-.');
grid on
```



Propriedades de sinais

Sinais periódicos

$$x(t) = x(t + kT), \forall t \in \mathfrak{R}$$

a) Verifique se o sinal $x(t) = \sin(t)$ é periódico.

Computando o período, obtém-se que $T = \frac{2\pi}{1} = 2\pi$. Analisando a periodicidade para $1 \leq k \leq 10$ e $1 \leq t \leq 5$.

```
t=1:5;
x=sin(t);
T=2*pi;
for k=1:10
    xk=sin(t+k*T)
end
```

```
xk = 1x5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1x5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1x5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1x5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1x5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
```

```

xk = 1×5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1×5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1×5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1×5
    0.8415    0.9093    0.1411   -0.7568   -0.9589
xk = 1×5
    0.8415    0.9093    0.1411   -0.7568   -0.9589

```

b) Construindo sinais periódicos

```

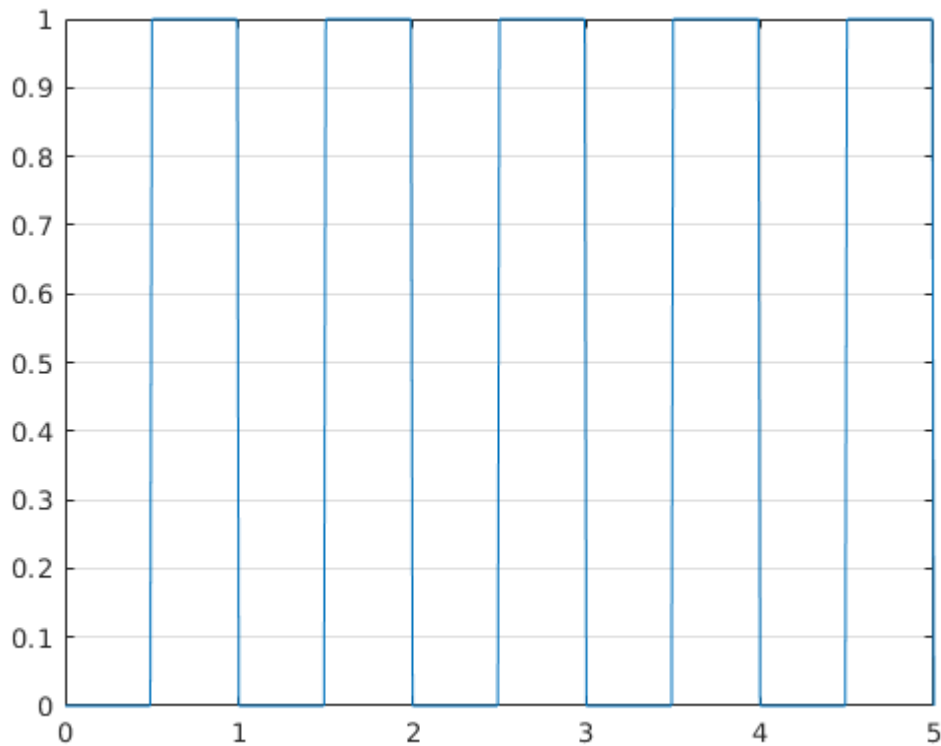
[s,t]=gensig(type,T,Tf,ts)
type: 'sin', 'square', 'pulse'

```

```

[s,t]=gensig('square',1,5,0.01);
plot(t,s)
grid on

```



```
square(t)
```

```

sawtooth(t,a)
% a é um escalar entre 0 e 1

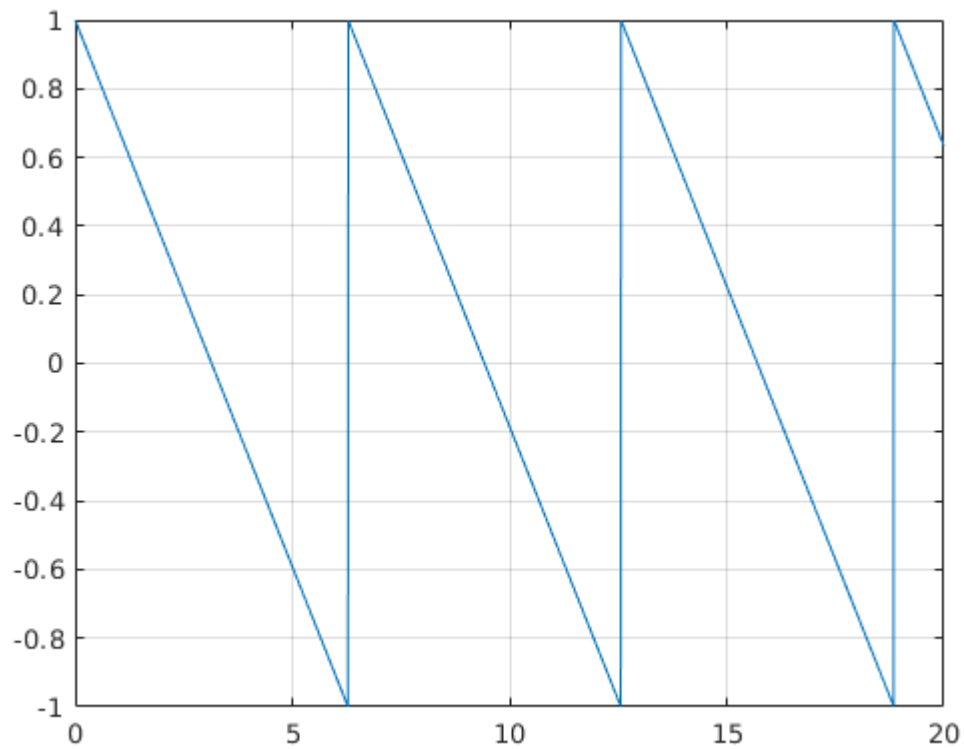
```

```

t=0:0.01:20;
sw=sawtooth(t,0);
plot(t,sw)

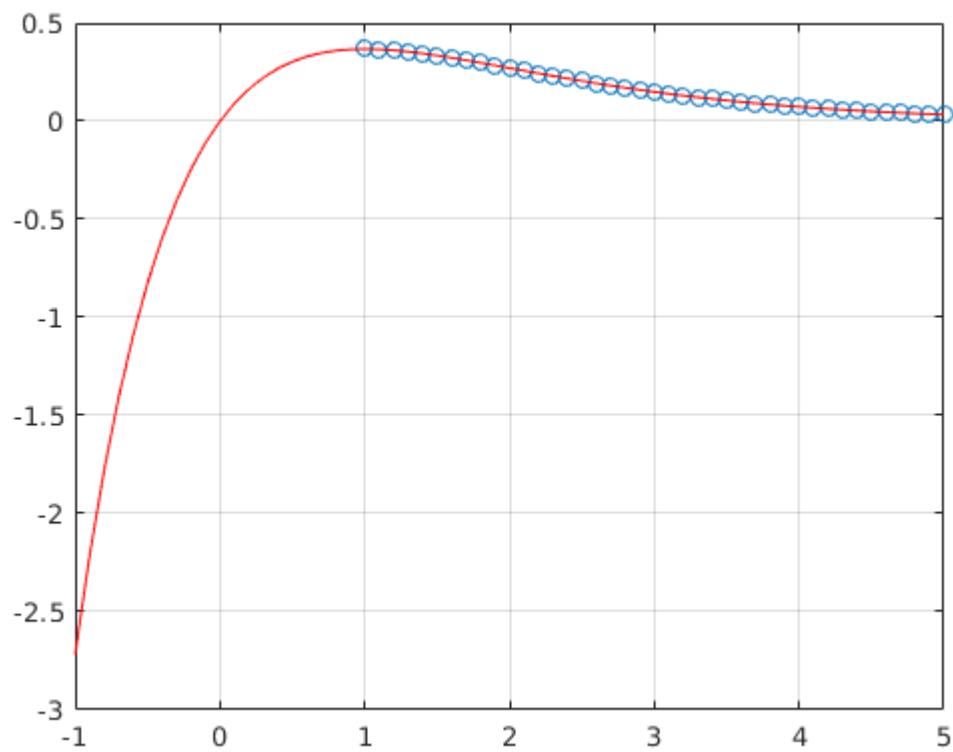
```

grid on



Sinais causais e não causais

```
t1=1:0.1:5;  
t2=-1:0.1:5;  
x1=t1.*exp(-t1);  
x2=t2.*exp(-t2);  
plot(t1,x1, '-o', t2,x2, '-r')  
grid on
```



Sinais do tipo par e ímpar

a) Verifique se os sinais $x(t) = t^2$ e $y(t) = t^3$ são do tipo par ou ímpar.

```
t1=0:1:4;
t2=0:-1:-4;
x1=t1.^2
```

```
x1 = 1x5
    0     1     4     9    16
```

```
x2=t2.^2
```

```
x2 = 1x5
    0     1     4     9    16
```

```
y1=t1.^3
```

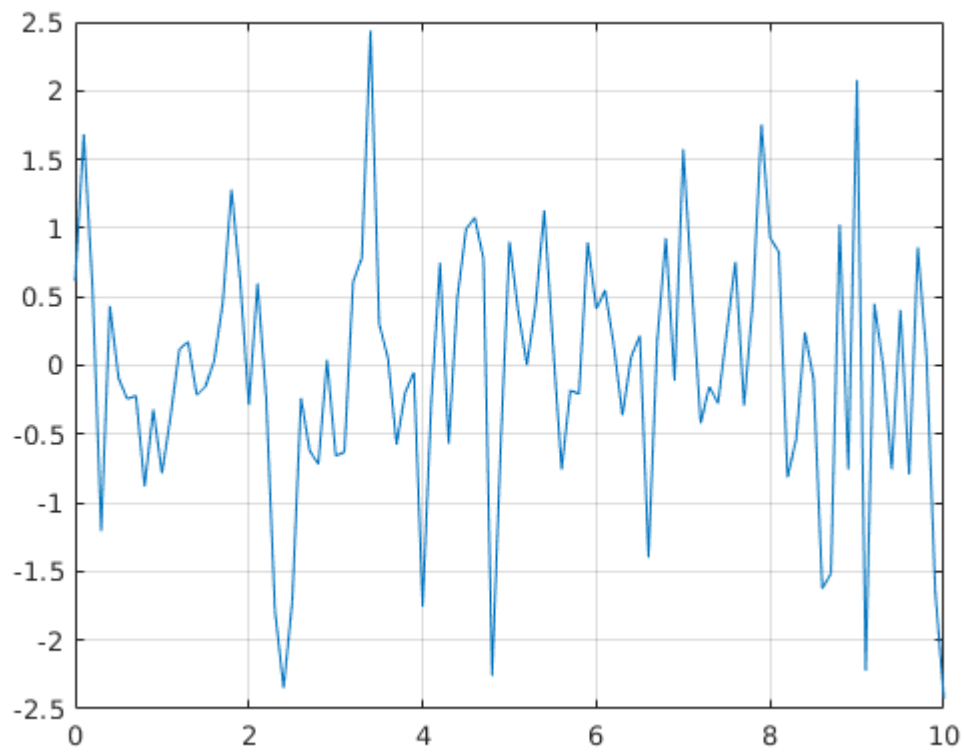
```
y1 = 1x5
    0     1     8    27    64
```

```
y2=t2.^3
```

```
y2 = 1x5
    0    -1    -8   -27   -64
```

Sinais determinístico e estocástico

```
t=0:0.1:10;
x1=randn(size(t));
plot(t,x1)
grid on
```



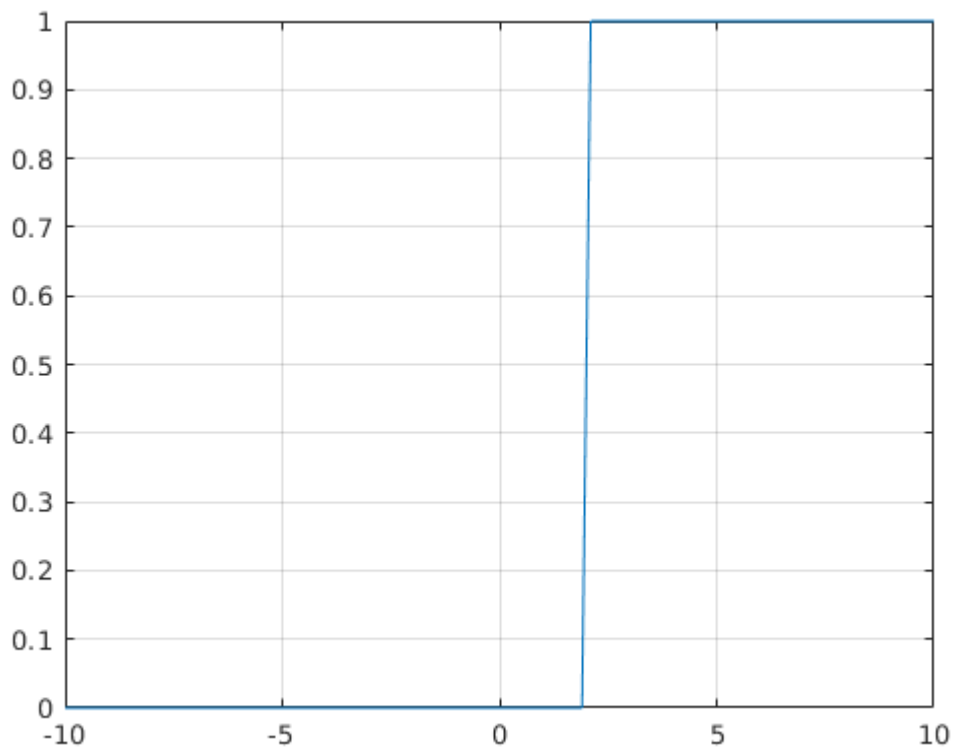
Transformação de sinais

Deslocamento temporal

a) Considere o sinal degrau unitário dado por $u(t - t_0)$, $t_0 = 2$. Obtenha a representação temporal de $u(t - 2)$.

Primeiro método

```
t=-10:0.1:10;
u=heaviside(t-2);
plot(t,u)
grid on
```



Segundo método

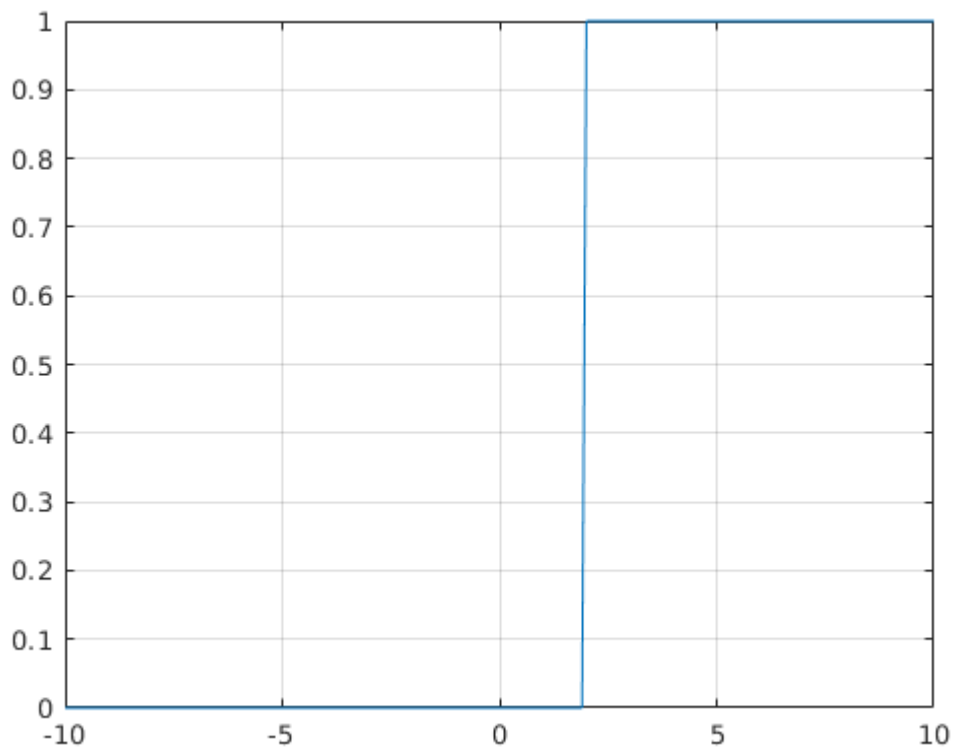
```
t1=-10:0.1:1.9;
t2=2:0.01:10;
u1=zeros(size(t1));
u2=ones(size(t2));
t=[t1 t2]
```

```
t = 1×921
   -10.0000   -9.9000   -9.8000   -9.7000   -9.6000   -9.5000   -9.4000   -9.3000 ...
```

```
u=[u1 u2]
```

```
u = 1×921
     0     0     0     0     0     0     0     0     0     0     0     0     0 ...
```

```
plot(t,u)
grid on
```

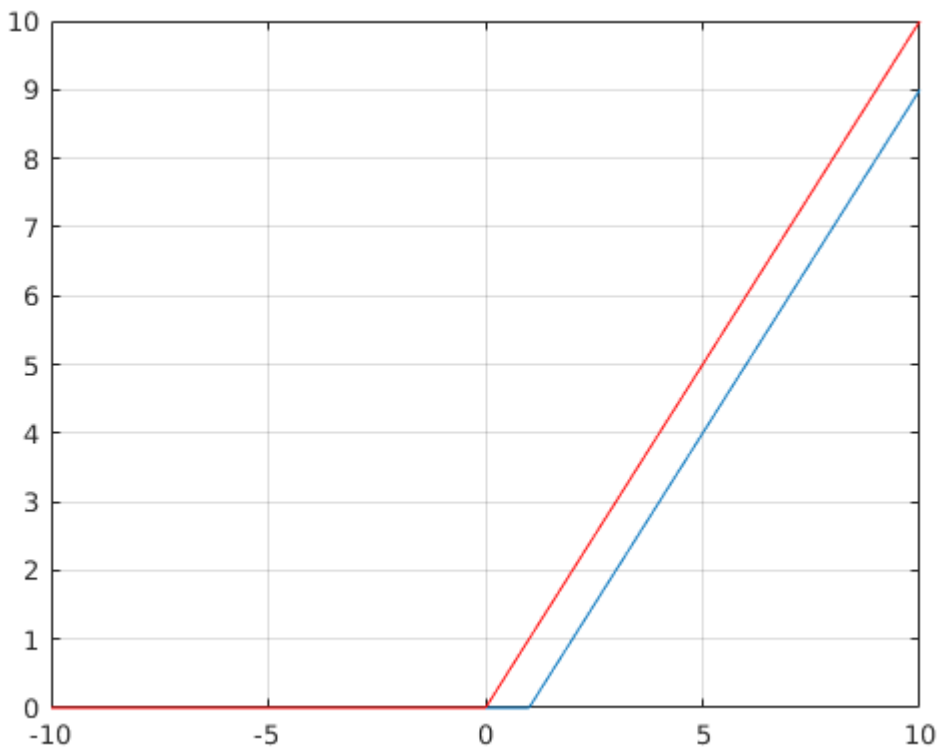


b) Obtenha a representação gráfica de $\delta(t + 2)$

c) Obtenha a representação gráfica da função rampa unitária, sendo $t_0 = 1$:

$$r(t - t_0) = (t - t_0)u(t - t_0) = \begin{cases} t - t_0 & t \geq t_0 \\ 0 & t < t_0 \end{cases}$$

```
t=-10:0.1:10;
r1=(t-1).*heaviside(t-1);
r2=(t).*heaviside(t);
plot(t,r1,t,r2,'r')
grid on
```

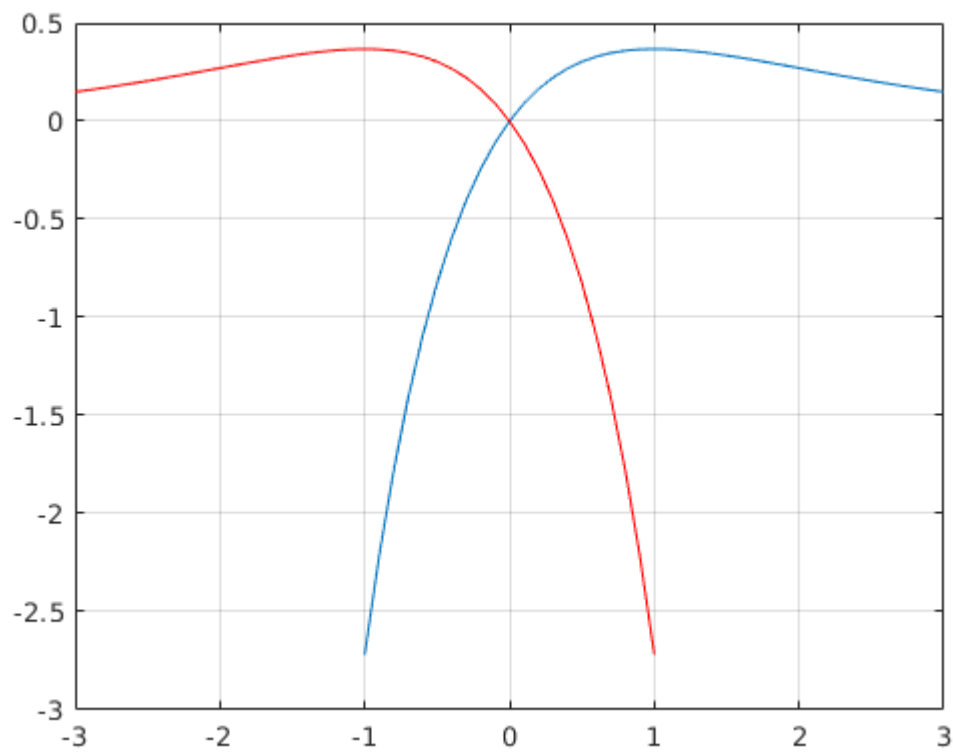



Reversão temporal ou reflexão

```
t=-1:0.1:3;  
x=t.*exp(-t)
```

```
x = 1×41  
-2.7183 -2.2136 -1.7804 -1.4096 -1.0933 -0.8244 -0.5967 -0.4050 ...
```

```
plot(t,x)  
hold on  
plot(-t,x, 'r')  
grid on
```



Escalonamento temporal

```
t=-1:0.1:3;
x=t.*exp(-t)
```

```
x = 1×41
    -2.7183    -2.2136    -1.7804    -1.4096    -1.0933    -0.8244    -0.5967    -0.4050 ...
```

```
figure
plot(t,x)
hold on

a=2;
plot((1/a)*t,x)

a=0.5;
plot((1/a)*t,x)
legend('x(t)', 'x(2t)', 'x(t/2)', 'location', 'bestoutside')
```

